ICE 409 — Mock Exam

Name: _____ Date: _____

97 multiple-choice items • 1 point each • No negative marking • Suggested time: 120 minutes (a practice default, not a professor-provided limit).

Choose the single best answer and record its letter. Use a separate sheet if needed. Do not open the answer key until you finish. Unless stated otherwise, calculations use the simplified lecture assumptions and consistent units; actual engineering work requires the applicable standard edition and full requirements.

Coverage follows the photographed pointers. Because the photo overlaps at item 81, this mock uses Part III 51–80 and Part IV 81–97. These are independently authored practice questions, not the actual exam. The full reviewer explains source corrections for dust zones/groups and unverified ISO revision claims.

Part I — Application | Questions 1–30

1. A field technician must confirm the exact power-supply terminals and DCS channel for LT-207. Which reference should be used first?

A. A high-level process flow diagram
B. The vessel material certificate
C. The project organization chart
D. The approved instrument loop diagram

2. An engineer is purchasing a transmitter for corrosive fluid. Which document should state its wetted materials and required accuracy?

A. The shift log alone
B. The cable tray loading schedule
C. The process overview alone
D. The instrument data sheet

3. A new high-high level trip is added. Which documentation response is most complete?

A. Revise only the filename of the P&ID
B. Keep the old trip matrix until the next shutdown regardless of risk
C. Change the PLC program without recording the change
D. Review the P&ID, I/O, loop, setpoint and cause-and-effect documents and revise all affected records

4. A pump may run only when level is adequate AND the discharge route is available. Both inputs are true, but an independent active-trip condition also blocks the pump. What is correct?

A. Assume a start request cancels every trip
B. Keep the pump blocked until the trip is resolved under the specified reset procedure
C. Delete the trip condition from the drawing
D. Start because two permissives are true

5. A technician needs the mounting bracket, process tapping, tubing and manifold arrangement for a pressure instrument. Which drawing is best?

A. The cybersecurity policy

B. The software version log

C. The alarm event list

D. The instrument hook-up drawing

6. A scenario has initiating frequency 0.4/year and one qualifying IPL with PFD 0.05. What is the frequency before adding the proposed SIF?

A. 8/year

B. 0.002/year

C. 0.45/year

D. 0.02/year

7. A scenario frequency before the SIF is 0.006/year and the tolerable frequency is 0.00002/year. What additional RRF is required?

A. 3,000

B. 300

C. 30

D. 0.00333

8. A low-demand safety function has calculated PFDavg of 0.0006. Which numerical band contains it?

A. SIL 2

B. SIL 3

C. SIL 1

D. SIL 4

9. A team credits a BPCS shutdown and an alarm as independent layers, but both depend on one transmitter whose failure initiates the scenario. What should the assessor do?

A. Reject the assumed independent multiplication and examine the shared failure

B. Ignore the initiating cause when evaluating independence

C. Increase the credited risk reduction because the hardware is shared

D. Multiply both credits automatically because their names differ

10. A HAZOP team applies the guide word Reverse to the parameter Flow. Which deviation should it study?

A. Flow in the unintended direction

B. Every software coding error regardless of process node

C. Flow at the intended direction and intended rate only

D. The total number of instruments on the site

11. Which assessment should be reconsidered when a low-demand function is changed to operate continuously?

A. Whether the site has a printer

B. The age of the cable schedule only

C. Use of PFDavg alone as the applicable quantitative metric

D. The spelling of the instrument tag alone

12. An independent high-high pressure shutdown is specified. Which arrangement represents the complete functional path?

A. A normal production setpoint alone

B. A safety PLC with no field inputs or outputs

C. Pressure sensor, safety logic solver and feed-isolation final element

D. Pressure display and a maintenance spreadsheet only

13. A SIL 2 target SIF is proposed with a SIL 3-capable PLC and an unassessed valve assembly. Can the target be claimed as achieved?

A. No; assess the complete function, including the final element and lifecycle requirements

B. Yes; the PLC capability automatically covers every connected device

C. Yes; only the highest component rating matters

D. Yes; valves never contribute dangerous failures

14. A 2oo3 pressure trip receives one high-pressure vote and two normal votes. Assuming no other trip conditions, is the voting threshold met?

A. No; all three votes are always required

B. Yes; the number 3 means any one channel

C. No; two of the three votes are required

D. Yes; one vote always satisfies 2oo3

15. A valve passes an online partial-stroke test. Which statement is justified?

A. It eliminated all future proof-test requirements

B. The test demonstrated the movement and failure modes covered by that procedure

C. It established zero dangerous-failure probability

D. It proved full shutoff tightness in every operating condition

16. A simplified 1oo1 subsystem has constant λDU = 2 × 10^-6/h and a 4,000-hour proof-test interval. Approximate PFDavg?

A. 0.04

B. 0.004

C. 0.008

D. 0.0005

17. A proof test identifies a stuck shutdown valve. What is the correct operational response?

A. Leave the bypass in place permanently
B. Call the test a pass because the PLC output changed
C. Manage the impairment, repair the fault, retest as required and document restoration
D. Erase the failure because the test found it

18. A safety software change alters the trip delay. What should happen before release?

A. Replace only the label on the panel
B. Review affected requirements and risk, control the change and verify/validate the affected function
C. Treat the change as cosmetic because the same PLC is used
D. Install it without recording the version

19. A proof test is complete, but an input bypass remains active. What must be done before claiming the SIF is available?

A. Remove the bypass under the controlled procedure and verify normal protection
B. Disable the bypass indication instead
C. Declare availability based solely on the signed test sheet
D. Assume the next process demand will clear the bypass

20. The control narrative says close the inlet valve on high-high level, but the cause-and-effect matrix says open it. What should the team do?

A. Average the two actions
B. Assume opposite actions are equivalent
C. Resolve the inconsistency through engineering review before implementing or accepting the function
D. Choose whichever document is easier to program

21. A flammable gas atmosphere is present continuously inside the assessed space. Which gas zone best describes this occurrence?

A. Zone 2
B. Zone 22
C. Zone 0
D. Zone 20

22. A dust cloud is unlikely in normal operation and lasts briefly if it occurs. Which zone applies to the assessed dust hazard?

A. Zone 22
B. Zone 0
C. Zone 2
D. Zone 20

23. Hydrogen is present in a classified area. A candidate device is approved only for gas group IIB. Which decision is appropriate?

A. Accept it because IIB is a dust group

B. Accept it because every Ex marking covers every gas

C. Do not assume suitability; verify equipment approval appropriate for IIC hydrogen service

D. Ignore the gas group if the enclosure is metal

24. A gas has autoignition temperature 120°C. Which listed rating satisfies the simple surface-temperature comparison?

A. T4: 135°C

B. T2: 300°C

C. T3: 200°C

D. T5: 100°C

25. An installer proposes a 24 V device without an intrinsic-safety assessment because it is low voltage. What is correct?

A. Ignore cable parameters in intrinsic-safety design

B. Treat a fuse alone as complete Ex i approval

C. Accept every 24 V circuit as intrinsically safe

D. Verify the complete certified circuit and associated apparatus; voltage alone is insufficient

26. An Ex d enclosure needs a replacement cable gland. What should determine the replacement?

A. Color match alone

B. Whether the gland can be forced into the hole

C. The cheapest available unmarked fitting

D. The enclosure certificate, cable construction and applicable protection/installation requirements

27. A 0–80 kPa linear transmitter produces 16 mA. What value does the signal represent?

A. 80 kPa

B. 60 kPa

C. 40 kPa

D. 16 kPa

28. A 0–50 m^3/h linear transmitter measures 30 m^3/h. What output is expected?

A. 12 mA

B. 16 mA

C. 9.6 mA

D. 13.6 mA

29. An analog cable must cross a power route. Which arrangement best follows the lecture guidance?

A. Remove the signal shield to avoid coupling

B. Use the required separation and cross near 90° where a crossing is necessary

C. Remove protective earth to reduce noise

D. Run both cables parallel in contact to reduce route length

30. All instruments power up, but no loop checks or interlock tests have been performed. What is the correct handover decision?

A. Complete required tests and integrated commissioning before operational acceptance

B. Accept the plant because power-up proves every function

C. Sign test records based on expected behavior

D. Treat neat wiring as proof of correct control action

Part II — Remembering | Questions 31–50

31. Which B31 code addresses liquid transportation pipelines?

A. B31.8
B. B31.4
C. B31.3
D. B31.1

32. What does MTR stand for in material control?

A. Material Test Report
B. Mechanical Timing Record
C. Motor Trip Relay
D. Maximum Temperature Ratio

33. Which ISO management standard focuses on occupational health and safety?

A. ISO 45001
B. ISO 14001
C. ISO 9001
D. ISO/IEC 27001

34. What does the first letter L normally identify in LT-305?

A. Low alarm only
B. Level
C. Logic solver
D. Local panel in every tag

35. What does a following R indicate in an instrument function such as TR?

A. Reset in every ISA tag
B. Radiation as the first variable in every position
C. Record
D. Ratio in every position

36. Which document lists cable identifiers, source, destination, type and core count?

A. Instrument data sheet only
B. Control narrative
C. Pressure test certificate
D. Cable schedule

37. Which document primarily shows overall major equipment and process streams?

A. Welder qualification record
B. Terminal-block diagram
C. Process Flow Diagram
D. Individual loop diagram

38. What does E/E/PE stand for?

A. Electrical, Electronic, Programmable Electronic
B. Emergency, Evacuation, Plant Exit
C. Equipment, Examination, Pipe Erection
D. Energy, Efficiency, Process Engineering

39. Which IEC 61508 part specifically addresses safety-related software requirements?

A. Part 3
B. Part 2
C. ISO Clause 7.5
D. ASME Section IX

40. What does ALARP stand for?

A. As Little As Regulations Permit
B. Always Leave A Risk Present
C. As Low As Reasonably Practicable
D. All Layers Automatically Remove Probability

41. What does HFT stand for?

A. Hazard-Free Technology
B. Hardware Fault Tolerance
C. High Frequency Transmission
D. Hardware Function Timing

42. Which item is normally a final element in a shutdown function?

A. Safety Requirements Specification document
B. Process variable label
C. Risk graph worksheet
D. Emergency shutdown valve and actuation assembly

43. What does BPCS stand for?

A. Basic Process Control System
B. Binary Programmable Cable Shield
C. Bypass Protection Confirmation Sheet
D. Boiler Pressure Certification Standard

44. Which term names the specific protective function implemented within an SIS?

A. SIF
B. PFD
C. SIL
D. MTR

45. Which metric is dimensionless?

A. PFDavg
B. Initiating-event frequency per year
C. PFH in failures per hour
D. Dangerous failure rate per hour

46. In NEC terminology, which class addresses flammable gases and vapors?

A. Class II
B. Temperature Class T6
C. Class III
D. Class I

47. Which corrected IEC dust group addresses nonconductive combustible dust?

A. IIIC
B. IIB
C. IIIB
D. IIA

48. What maximum surface temperature corresponds to T6?

A. 100°C
B. 135°C
C. 450°C
D. 85°C

49. What does LSZH describe in cable materials?

A. Line Shield Zone Hazard
B. Liquid-Sealed Zinc Housing
C. Low Smoke Zero Halogen
D. Low Signal Zero Harmonics

50. What does PE denote in the installation material?

A. Protective earth
B. Pressure examination
C. Programmable efficiency
D. Process estimate

Part III — Application | Questions 51–80

51. A steam-generation piping system requires selection from the listed B31 codes. Which is the primary power-piping reference?

A. ISO 45001
B. B31.4
C. B31.1
D. B31.8

52. A process line operates safely, but the proposed design omits an anticipated pressure surge. What is the correct response?

A. Accept the omission if the paint is intact
B. Evaluate the surge in the applicable design-pressure/load assessment
C. Use a tag number instead of a pressure calculation
D. Ignore it because only average pressure matters

53. Pressure-design inputs are P = 1 MPa, D = 100 mm, S = 100 MPa, E = 1 and Y = 0.4. Using the lecture formula, what is t?

A. Approximately 0.498 mm
B. Approximately 1.004 mm
C. Approximately 50 mm
D. Approximately 4.98 mm

54. Pressure thickness is 3 mm and corrosion allowance is 2 mm. Negative mill tolerance is 12.5%. Minimum nominal thickness from these factors?

A. Approximately 5.714 mm
B. Approximately 4.375 mm
C. 5 mm
D. Approximately 3.429 mm

55. A material is proposed at a higher metal temperature than originally specified. What should be rechecked in the wall-thickness calculation?

A. Only the instrument-loop number
B. The applicable allowable stress and other temperature-dependent factors
C. Only the shipping weight
D. Only the pipe color

56. An unlisted material has no permitted qualification evidence. What is appropriate?

A. Accept it because it is expensive
B. Assume it has the same allowable stress as any steel
C. Use a larger label to compensate
D. Require the code-permitted material evaluation before acceptance

57. A welder has qualifications, but the proposed welding procedure lacks supporting qualification. What follows?

A. Accept any procedure used by that welder
B. Review and establish the required procedure qualification; personnel qualification alone is insufficient
C. Treat personnel and procedure qualification as interchangeable
D. Skip the WPS if an inspector is present

58. A joint requires PWHT under the applicable rules. The contractor proposes to omit it to shorten the schedule. What should happen?

A. Let the schedule determine metallurgical acceptance
B. Follow the applicable PWHT requirement or an explicitly permitted engineered alternative
C. Omit it because a visual inspection passed
D. Replace it with a new revision number alone

59. An owner wants to confirm that examination records and welding certifications satisfy the construction requirements. Which activity is this?

A. Only purchasing a new valve
B. Only drawing a cable route
C. Only measuring process flow
D. Inspection of compliance and records

60. A 4 MPa design line is proposed for a pneumatic test. What lecture multiplier gives an initial study value before required adjustments and limits?

A. $1.1 \times 4 = 4.4$ MPa
B. $1.5 \times 4 = 6$ MPa automatically
C. $4 \times 4 = 16$ MPa
D. $0.1 \times 4 = 0.4$ MPa

61. A trapped liquid section can overpressure as it warms. Which protection addresses that scenario?

A. A new instrument tag with no hardware change
B. A cable shield disconnected at both ends
C. A larger display font
D. An appropriately selected thermal relief arrangement

62. A process pipe has measurable in-service corrosion. Which listed assessment addresses continued suitability of damaged equipment?

A. Only an ISA tag-decoding exercise
B. Only the original paint specification
C. Only a supplier attendance list
D. Fitness-for-service assessment using the applicable guidance

63. A manager defines accuracy targets, resources and a calibration schedule. Which PDCA phase is being emphasized?

A. Do
B. Act
C. Plan
D. Check

64. Technicians perform the approved calibration procedure. Which PDCA phase is most directly represented?

A. Plan
B. Check only
C. Act only
D. Do

65. An engineer compares calibration failure rates with the quality target. Which PDCA phase is most directly represented?

A. Do only
B. Check
C. Procurement outside PDCA
D. Plan only

66. A recurring wiring error leads to an improved installation procedure, training and later effectiveness review. What is being demonstrated?

A. An automatic exemption from future QC
B. Abandonment of quality planning
C. Corrective action and continual improvement
D. A correction with no cause analysis

67. A team audits a supplier before awarding an instrumentation order. Which classification best fits?

A. Quality assurance focused on process confidence
B. Only final-product quality control
C. A pressure proof test
D. A hazardous-area zoning method

68. A delivered pressure transmitter is tested against its specified accuracy limits. Which classification best fits?

A. Only relationship management
B. Quality control of the output
C. Only management leadership
D. Only an environmental assessment

69. A known supply delay threatens commissioning. Which action best applies risk-based thinking?

A. Hide the delay in the audit log
B. Wait until the deadline is missed before identifying the issue
C. Assess the risk early and plan mitigation and contingencies
D. Eliminate traceability requirements to save time

70. An audit reveals outdated drawings in field folders. Which corrective approach is best?

A. Fix distribution/access control and verify users receive current approved revisions
B. Tell workers to guess which revision is newest
C. Change the title without reviewing its content
D. Delete all controlled master documents

71. A drawing shows PT-112 and PIC-112. Which valve tag is consistent with the lecture's pressure-loop convention?

A. TV-112
B. LV-112
C. FV-999
D. PV-112

72. The same letter I appears first in an instrument tag and later as a functional letter in another. What is the correct interpretation?

A. I always indicates an intrinsically safe circuit
B. I always means the instrument is indoors
C. I always means an interlock regardless of position
D. First I can mean electrical current; following I means indication

73. An operator sees LSHH-401. What should the tag be read as?

A. Low-speed heating controller, loop 401
B. A transmitter that must output 401 mA
C. Level sensor heat-humidity, loop 401
D. Level switch high-high, loop 401

74. An instrument bubble has no location line in the lecture's chart. Which interpretation is appropriate?

A. Normally field mounted and accessible at the field location
B. Guaranteed SIL 4 certification
C. Guaranteed primary-panel mounting
D. Always behind the main panel

75. A circle has a dashed horizontal location line in the supplied chart. What does it normally communicate?

A. A function behind the main panel, normally not operator accessible

B. An unconditionally invalid symbol

C. A failed instrument battery

D. A broken pressure pipe

76. A fuel valve must stop flow on loss of the specified actuator utility. Which failure action fits that defined safe state?

A. Fail open

B. Fail closed

C. Always drift open after a delay

D. Fail indeterminate

77. A valve is labeled FL/DO. Which behavior must be considered after the specified loss of power or air?

A. It always reaches exactly 50% travel

B. It initially holds its last position and can later drift open

C. It is a certified pressure relief valve

D. It immediately and permanently locks closed

78. In the reactor example, a 4–20 mA controller output is converted to a 3–15 psi actuator signal. What function is represented?

A. Direct proof of complete-loop SIL 3

B. Conversion of process steam into cooling water

C. Conversion of pressure piping into a data bus

D. Electrical-to-pneumatic signal conversion

79. A downstream chlorine analyzer detects residual below its setpoint in the supplied dosing example. What is the intended normal control response, within specified constraints?

A. Shut every unrelated cooling valve

B. Always reduce dosing further

C. Increase chlorine dosing toward the target

D. Assume an analyzer cannot be part of feedback control

80. A P&ID includes a dashed electrical signal between a transmitter and controller. What does that connection primarily represent?

A. Main flow of process fluid

B. The exact physical routing dimension

C. Transmission of measurement/control information

D. A weld requiring PWHT

Part IV — Understanding | Questions 81–97

81. Why is a Zone 2 location still subject to hazardous-area equipment selection?

A. Zone 2 is an unclassified office by definition
B. Zone 2 guarantees no flammable material exists
C. Only locations with continuous explosions are hazardous
D. An explosive atmosphere can still arise under the specified abnormal conditions

82. Why should gas Zones 0/1/2 and dust Zones 20/21/22 not be mixed?

A. They describe different hazardous-atmosphere types using separate zone series
B. Gas zones identify cable conductor size
C. Dust can never cause an explosion
D. The numbers are interchangeable spelling choices

83. How do Ex i and Ex d differ in basic principle?

A. Ex i limits ignition-capable energy; Ex d contains an internal explosion and prevents flame transmission
B. Both rely only on a high IP number
C. Ex i contains unrestricted internal explosions; Ex d eliminates every electrical energy source
D. Both make the gas group irrelevant

84. Why is T6 cooler than T4 despite having a larger class number?

A. The number is the minimum ambient temperature
B. T6 is hotter because six is larger than four
C. The class number is a label; its permitted maximum surface temperature is lower
D. The rating is the number of heating elements

85. Why can combustible dust deposits matter even without a continuous airborne cloud?

A. Deposits automatically change every area to gas Zone 0
B. Deposits can insulate hot surfaces and may later become airborne
C. Dust deposits are always inert after settling
D. Deposits remove the need for temperature assessment

86. Why does an IP rating alone not establish hazardous-area suitability?

A. A numeric enclosure rating eliminates certification conditions
B. Any watertight enclosure is automatically flameproof
C. Ingress protection and explosion protection address different requirements
D. IP always means intrinsic protection with universal gas approval

87. What is the primary difference between cable armor and shielding?

A. Armor alone proves a circuit intrinsically safe
B. Shielding is a substitute for all conductor insulation
C. Armor primarily resists mechanical damage; shielding primarily manages electromagnetic interference
D. They are always the same layer with identical performance

88. Why is protective earth not an acceptable conductor to remove when diagnosing signal noise?

A. It is only a spare communication wire
B. It carries no safety function if a signal is 4–20 mA
C. It performs an electrical-safety function that must remain effective
D. Its removal automatically increases measurement accuracy safely

89. Why is a single-point shield connection common for many low-frequency analog loops but not universal?

A. Shield connections have no effect on interference
B. A shield can never be connected at a panel
C. Every network uses identical grounding requirements
D. Ground-loop and high-frequency EMC requirements differ between systems

90. How does commissioning differ from a visual installation inspection?

A. Commissioning eliminates every need for calibration
B. Commissioning demonstrates integrated operational behavior; inspection checks installation conformity
C. Inspection automatically proves all control sequences
D. Both are only checks of paint finish

91. Why can a correctly calibrated transmitter still produce an incorrect DCS display?

A. A correct local reading proves every signal terminal
B. DCS displays cannot be configured
C. Wiring, channel assignment or scaling elsewhere in the installed loop can be wrong
D. Calibration guarantees every connected software setting

92. Why should an instrument keep the same identification across the index, P&ID and loop diagram?

A. Different tags always make maintenance safer
B. Only the drawing title matters
C. Consistent identity supports traceability and prevents installation/troubleshooting errors
D. Matching tags remove the need for technical specifications

93. Why does redundancy not automatically remove systematic failure risk?

A. Systematic errors occur only in cable insulation
B. Redundant channels may share the same specification, software or design error
C. Two channels cannot have common causes
D. Any voting scheme guarantees zero risk

94. Why is a risk graph not a universal SIL answer based on consequence alone?

A. Its calibrated decision path also considers exposure, avoidance and occurrence assumptions
B. Its output depends only on the color of the diagram
C. Consequence is never relevant
D. Every hazard in every plant requires the same SIL

95. Why must proof-test interval and coverage be included in SIL verification?

A. Proof-test coverage is automatically 100% for every procedure

B. They influence how long dangerous hidden faults can persist and which faults are revealed

C. Testing intervals are always identical for all devices

D. They are unrelated to failure probability

96. How does SIS purpose differ from BPCS purpose?

A. The SIS performs required protection; the BPCS performs routine process regulation

B. The SIS only optimizes production cost

C. Both names guarantee identical independence and integrity

D. The BPCS is always a complete independent safety layer for every scenario

97. Why is verification different from validation?

A. Verification is only painting the cabinet; validation is only printing a certificate

B. Verification checks outputs against requirements; validation confirms the integrated function fulfills its intended safety requirements

C. Validation is unnecessary if a device has a serial number

D. They both mean selecting the cheapest available component